Manchester.Ai_IBM Z Datathon 2026 Project

Theme 1: Real-Time AI for Critical Decisions · Event code SSF2026

StudentShield

Real-time money-mule early warning on IBM LinuxONE: warn the student before the money moves on

Thinojen Balakumaresan · Julian Newson · Sshubhan Kammari · Arda Yencer · University of Manchester

Young people are recruited as money mules, and caught only after the money has gone

23%

of suspected mule cases on the Cifas database involved people aged 21 and under

Cifas & UK Finance, H1 2023

64%

involved people aged up to 30

Cifas & UK Finance, H1 2023

£1.28bn

lost to UK payment fraud in 2025

UK Finance, Annual Fraud Report 2026

Usual approach: the bank finds the mule after the chain is complete. The victim's money is gone; the student loses their account and may be prosecuted.

Our approach: act in the hours between money landing and money leaving.

IBM's own AML data, evaluated the way a bank would

IBM Transactions for AML, HI-Small (synthetic, fully labelled): 5.08M payments, 515k accounts, 1 in 1,000 laundering, 15 currencies converted to USD

Warm-up	Train	Validation	Test
1 Sep history only	2–6 Sep 2.09M payments	7–8 Sep 945k · cut-offs chosen	9–10 Sep 845k · touched once

1. **Split by date, never randomly**
   A bank trains on last month and scores next month. A random split would leak mule rings into the test set.

2. **Point-in-time features**
   All 25 features use only payments strictly before the one being scored: exactly what the bank knows when money lands. Checked against brute force.

3. **Honest cleaning**
   Dropped 11 Sep onwards (the generator stops making normal payments) and self-transfers (11.6% of rows, almost no laundering).

How it works: one score, two people told

01

Payment lands

Each incoming payment is sent to the scoring service

→

02

25 signals

The payment, the receiving account (fan-in, pass-through, age) and the sender (fan-out)

→

03

XGBoost score

Scored on LinuxONE in under a millisecond

→

04

Reasons

Top 3 SHAP contributions become plain-English reasons

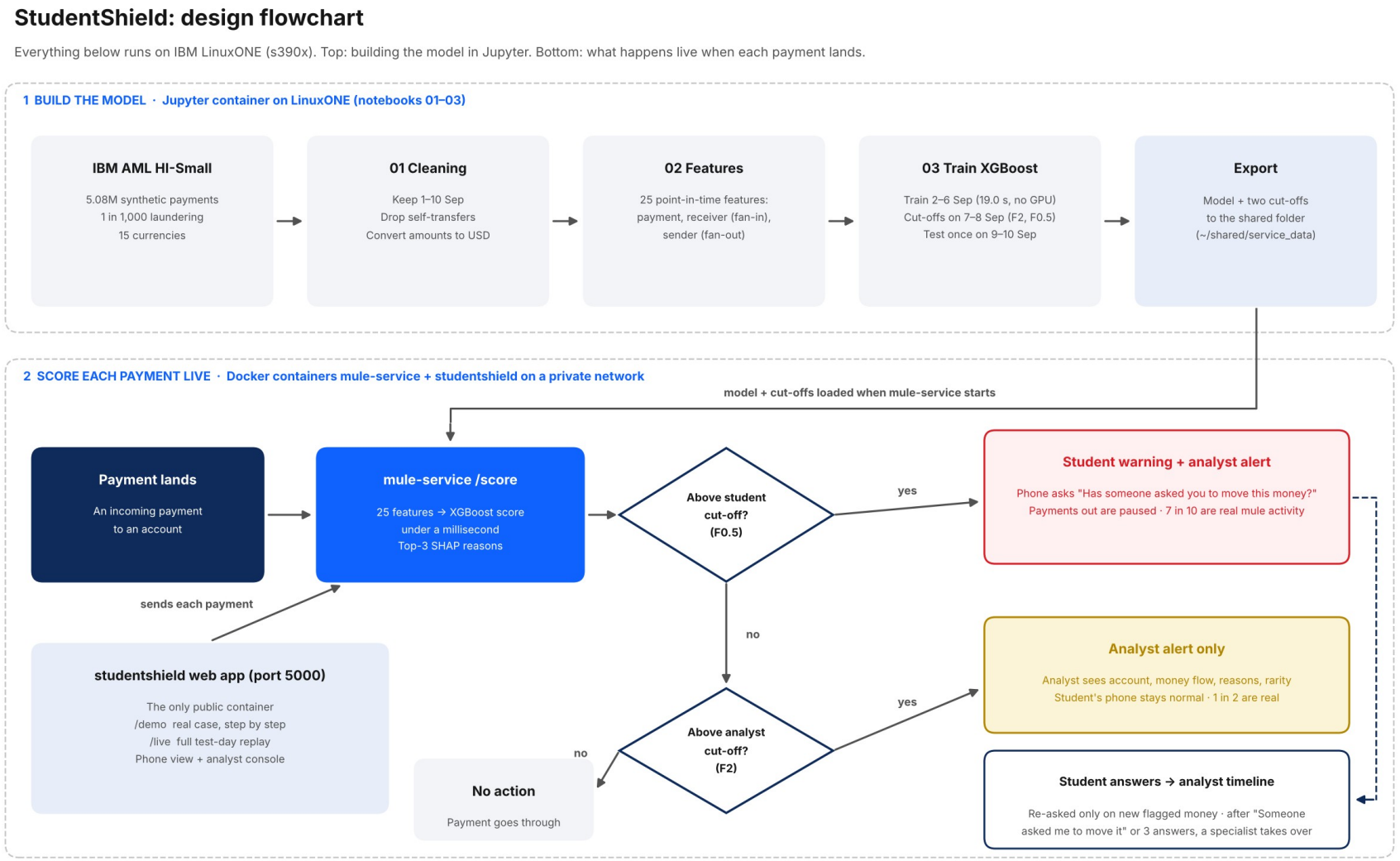
Analyst alert

Recall-weighted cut-off (F2). The bank's analyst sees the account, money flow, reasons and rarity ("1 in N payments scores this high"). The student's phone stays normal.

Student warning

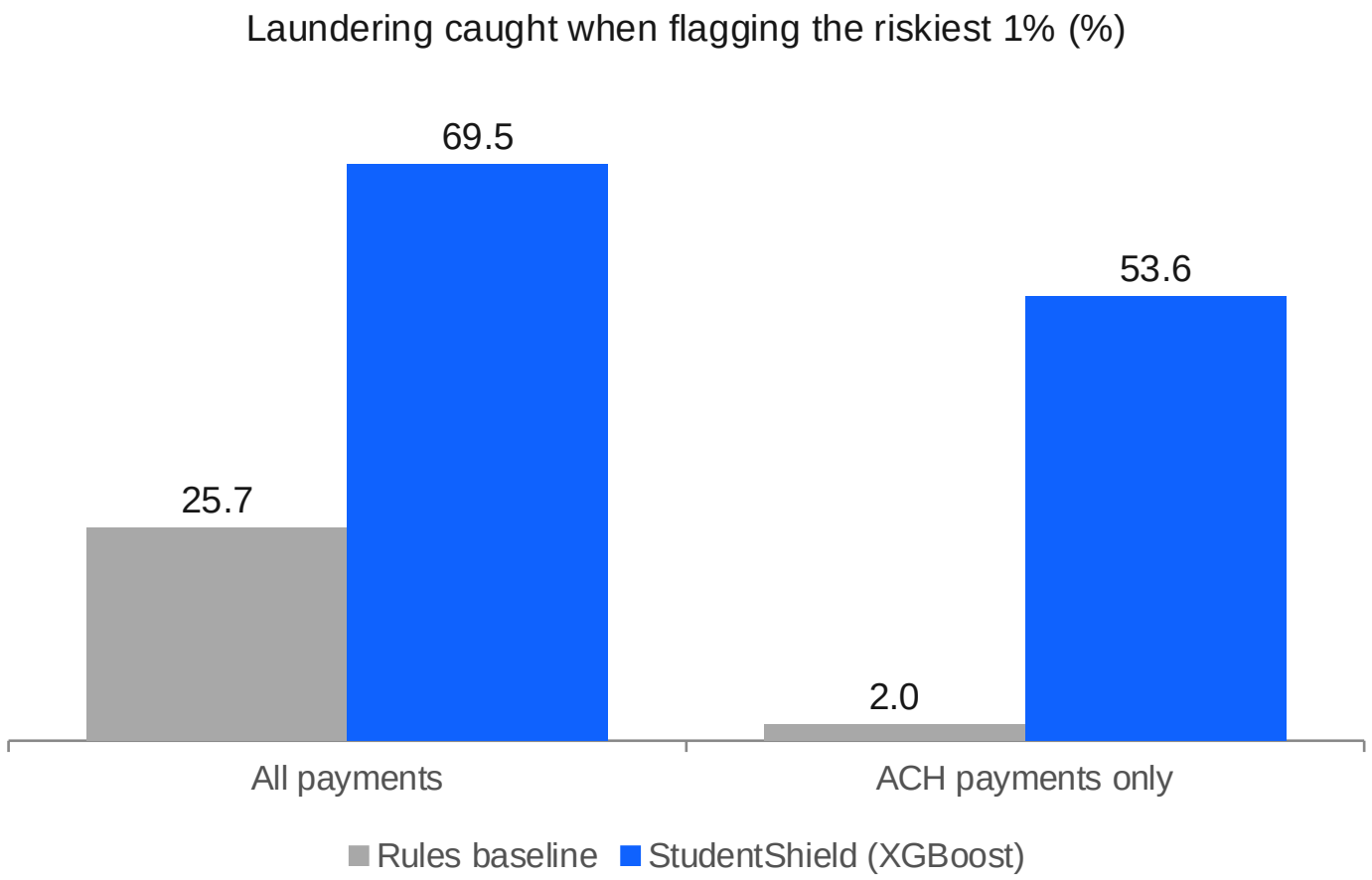
Precision-weighted cut-off (F0.5). A calm question on the phone: "Has someone asked you to move this money?" Payments out are paused and every answer goes to the analyst.

Design flowchart



2.7× more laundering caught than a bank-style rule

Test days 9–10 Sep: ~845k payments the model never saw, at the real 0.1% laundering rate



0.430

PR-AUC, vs 0.024 for the rules (18×)

53.6%

caught within ACH alone, where the rules collapse to 2.0%: it learns mule behaviour, not payment type

19.0 s

to train on 2.09M payments on LinuxONE, no GPU

7 in 10 student warnings are real mule activity

Each cut-off maximises F-beta on validation, then is measured once on test. The bank can move along the trade-off curve.

Analyst alert (F2)

49.5% of alerts are real (1 in 2)

45.2% of all laundering caught

~220 false alarms a day, reviewed by a person

Student warning (F0.5)

69.7% of warnings are real (7 in 10)

37.8% of all laundering caught

~78 wrong warnings a day: about 1 in 5,400 innocent payments

Live demo: a real case from the unseen test days

http://148.100.108.113:5000 · every score on screen is real model output on LinuxONE

Payment stream

Real payments arrive; account tiles turn amber at analyst level and red at student level

Student's phone

Warning, "Has someone asked you to move this money?", outgoing payments paused, re-asked only when new flagged money arrives

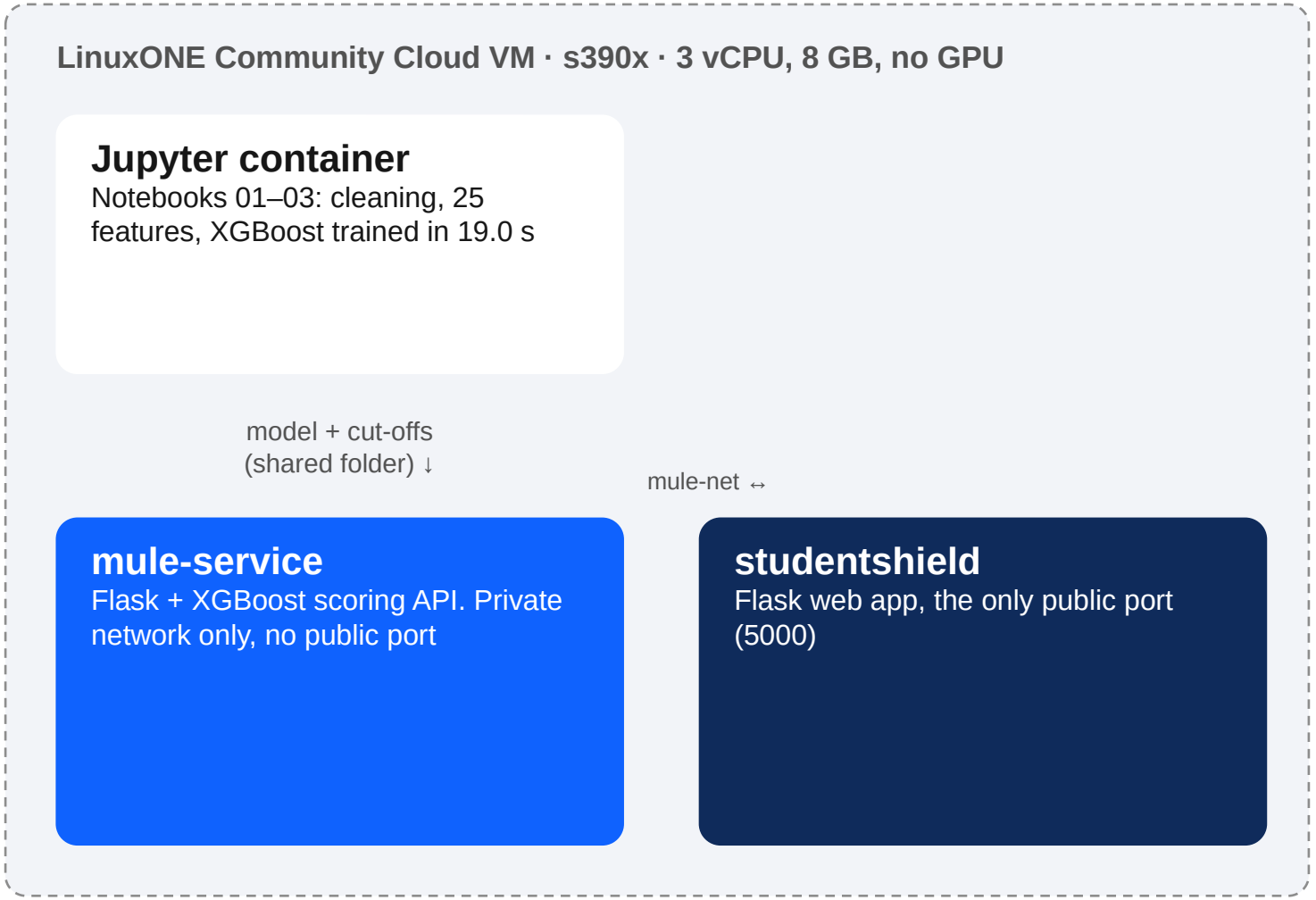
Analyst console

Rarity, money-flow graph with the held transfer, three plain-English reasons, and every answer the student gave

~19 h

In the case played, the student was warned about 19 hours before the money moved out, and the outbound transfer was held.

Built and running end to end on IBM LinuxONE (s390x)



On-platform AI
Training and scoring both run on s390x; payment data never leaves the platform it lives on

Containerised
Docker images on IBM's s390x base image, --restart unless-stopped: both came back on their own after a VM restart

Secure by design
The model API sits on a private Docker network; only the web app is exposed

Real-time
Under a millisecond per payment; stateless scoring scales out with more replicas

Designed to protect students, not accuse them

No tipping off
The phone never shows a score, reason or suspicion: only a safeguarding question banks already ask

A person decides
The model never freezes an account or files a report; an analyst reviews every alert

Strict cut-off for people
Students are contacted only at the precision-weighted level; holds happen only there

No identity features
No age, nationality or ethnicity in the model; currency features flagged for a fairness check before real use

Honest evaluation
Synthetic data labelled as such; test set used once; the failed Isolation Forest result reported

Privacy
Synthetic data, no real customers; scoring stays on the platform, nothing sent to outside services

A smoke alarm, not a fire investigation

Code: github.com/ksshubhan/manchester-imbz-datathon-2026 · Live: http://148.100.108.113:5000

Open source: XGBoost (xgboost.ai) · pandas (pandas.pydata.org) · NumPy (numpy.org) · scikit-learn (scikit-learn.org) · Flask (flask.palletsprojects.com) · JupyterLab (jupyter.org) · Docker (docker.com)
Data: IBM Transactions for AML, HI-Small (Kaggle; Altman et al., NeurIPS 2023)

Thinojen Balakumaresan · Julian Newson · Sshubhan Kammari · Arda Yencer